

# FreshFlow Beverages Pvt. Ltd.

FreshFlow Bottling Plant - Pune Production Facility  
FSSAI: 10012022000345 | ISO 22000:2018 / ISO 14001:2015

## STANDARD OPERATING PROCEDURE

### Shutdown Procedure for Mixing Line 62

Department: Mixing

|                                                  |
|--------------------------------------------------|
| Document Number: SOP-062                         |
| Revision: Rev 1.0                                |
| Classification: CONFIDENTIAL - INTERNAL USE ONLY |
| Prepared by: Engineering / QA Department         |
| Approved by: Plant Manager                       |

## Document Control Information

| Parameter        | Value                                               |
|------------------|-----------------------------------------------------|
| SOP Number       | SOP-062                                             |
| Title            | Shutdown Procedure for Mixing Line 62               |
| Department       | Mixing                                              |
| Plant            | FreshFlow Bottling Plant - Pune Production Facility |
| Prepared By      | Employee 4                                          |
| Reviewed By      | Rajesh Kulkarni                                     |
| Approved By      | Rajesh Kulkarni - Plant Manager                     |
| Issue Date       | 2023-03-15                                          |
| Next Review Date | 2026-06-19                                          |
| Revision         | Rev 3.3                                             |

## 1. Purpose

Blend water, syrup, and CO2 to create a finished carbonated beverage product meeting Brix, pH, CO2, and microbiological specifications.

## 2. Scope

Mixing department operators.

## 3. Prerequisites and Requirements

### 3.1 Training Requirements

- Mixing and carbonation training
- CO2 safety training (SOP-SAF-001)

### 3.2 Tools and Materials Required

- Calibrated instruments as listed in the plant calibration schedule.
- Appropriate PPE as specified in Section 4.
- Completed permit-to-work (if required for maintenance activities).
- Access to relevant equipment manual and P&ID drawings.
- CMMS access to create and close work orders.

## 4. Health, Safety and Environmental Requirements

### ! WARNING

This SOP must not be carried out without appropriate PPE and, where required, a valid Permit to Work (SOP-SAF-006). If in doubt about safety, STOP and consult the Safety Officer (EMP009 / EMP017).

### 4.1 Identified Hazards

- CO2 (asphyxiation risk at concentrations >5% vol) - monitor GD101
- Cold product at 2-4°C - skin hazard and slip risk
- Pressurized systems (up to 6 bar CO2) - release risk

### 4.2 Emergency Response

In case of any emergency during this procedure: Press the nearest Emergency Stop, alert personnel in the vicinity, call the Safety Officer and Shift Supervisor. Refer to the Emergency Response Plan (SOP-SAF-004) and site evacuation procedure.

## 5. Detailed Procedure

### 5.1 Pre-Task Verification

1. Confirm the relevant equipment is available and no conflicting activities are planned.
2. Check shift handover log for any outstanding issues with this equipment.
3. Verify all required materials, tools, and chemicals are available and correctly labelled.
4. Obtain all required permits (PTW, hot work, confined space as applicable).
5. Brief all involved personnel on this SOP before starting.

### 5.2 Main Procedure

1. Notify the shift supervisor and next upstream/downstream department of planned shutdown.
2. Reduce production rate to minimum before initiating shutdown.
3. Initiate shutdown sequence from SCADA or local panel as per control configuration.
4. Allow equipment to reach rest before applying isolation.
5. Close all isolation valves (suction, discharge, utility) in the sequence specified in the P&ID.
6. For food-contact equipment: initiate CIP sequence before shutdown if required.
7. Apply LOTO if shutdown is for maintenance purpose - see SOP-SAF-002.
8. Record shutdown time, reason, and condition of equipment in the shift log.